Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		• Hadrons are made up of quarks • Quarks / hadrons feel the strong force. • Hadrons can be split into two groups, baryons and mesons [Accept three groups and reference to anti-baryons] • Baryons contain three quarks. Protons and neutrons are baryons. • Protons contain 2 up quarks and 1 down quark. • Neutrons contain 1 up quark and 2 down quarks. • Up quarks carry a charge of $+\frac{2}{3}$, down quarks carry a charge of $-\frac{1}{3}$. • For protons: $+\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1$ For neutrons: $+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$. • Mesons contain a quark and an antiquark or detail e.g π^+ is $\mathrm{u}\bar{\mathrm{d}}$ $(+\frac{2}{3}+\frac{1}{3})$ etc • There are 3 generations of hadrons / quarks. • Hadrons have a lepton number of 0, baryons have a baryon number of 1, anti-baryons -1 and mesons 0. **5-6 marks** A comprehensive analysis of baryons and mesons. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** A comprehensive account of either baryons or mesons or a limited account of both. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.*	6			6		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			1-2 marks A limited account of either baryons or mesons. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** *No attempt made or no response worthy of credit.*						
	(b)		Charge: +1 -1 = 0 +1 -1 (1) Baryon: 1 + 0 = 1 + 0 + 0 (1) Accept: $\mathrm{uud} + \bar{\mathrm{u}}\mathrm{d} = \mathrm{udd} + \mathrm{u}\bar{\mathrm{d}} + \bar{\mathrm{u}}\mathrm{d}$ Lepton: 0 + 0 = 0 + 0 + 0 (1) N.B. Charge and lepton number must be present and either baryon or quark number		3		3		
			Question 5 total	**6**	**3**	**0**	**9**	**0**	**0**